R13

Code No: 115EH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) What system calls have to be executed by a command interpreter or shell in order to start a new process? Explain briefly.
b) Explain briefly about virtual machines and micro Kernels. [8+7]

2.a) What are the major activities of an operating system with regard to file management? Explain them briefly with their supporting system calls.
b) Discuss various operating system services that are very much helpful to the user. [8+7]

3. What is a monitor? Explain how dinning philosopher's problem is solved using monitors with example pseudo code. [15]

4.a) What resources are used when a thread is created? How do they differ from those used when a process is created?
b) Discuss readers/writers problem and give solution by using semaphores where writers have priority. [7+8]

5. Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 KB, 4 KB, 20 KB, 18 KB, 7 KB, 9 KB, 12 KB, and 15 KB. Which hole is taken for successive segment requests of: (a) 12 KB (b) 10 KB (c) 9 KB for first fit, best fit, worst fit, and next fit approaches. [15]

6.a) Explain briefly about LFU Page replacement algorithm.
b) Give the detailed description of hardware implementation of a page table with translation Look-Aside Buffer. [7+8]

7. Explain the following with relevant diagrams:
a) Two level directory structure.
b) Acyclic-graph directory structure. [7+8]

8. List the conditions that must be present for deadlock to occur and for each condition, give brief example or reason that illustrate a disadvantage in preventing the condition. [15]

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